

Advanced (Habanero)

- Q1.** Evaluate $f(g(x))$ if $f(x) = x^2$ and $g(x) = 2x + 1$
(A) $4x^2 + 4x + 1$
(B) $4x^2 - 4x - 1$
(C) $4x^2 + 4x - 1$
(D) $2x^2 + 2x + 1$
(E) $2x^2 - 2x - 1$
- Q2.** Find the domain of $f(g(x))$ where $f(x) = \frac{1}{x}$ and $g(x) = x^2 - 4$
(A) $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$
(B) $(-\infty, -2) \cup (2, \infty)$
(C) $(-2, 2)$
(D) $[0, \infty)$
(E) $(-\infty, -2) \cup [2, \infty)$
- Q3.** Interpret $f(g(x))$ in the context of a population growth model, where $f(x) = 2x$ and $g(x) = x + 5$
(A) The population doubles every 5 years
(B) The population increases by 5 every year
(C) The population doubles every year, and then 5 is added
(D) The population increases by 10 every 5 years
(E) The population increases by 5, then doubles
- Q4.** Given $f(x) = |x|$ and $g(x) = x - 3$, find $f(g(2))$
(A) 1
(B) 3
(C) 5
(D) -1
(E) -3
- Q5.** Determine the range of $f(g(x))$ where $f(x) = x^2$ and $g(x) = \sqrt{x}$
(A) $[0, \infty)$
(B) $(-\infty, 0]$
(C) $[0, 1]$
(D) $(-1, 1)$
(E) $[1, \infty)$
- Q6.** Find the value of x for which $f(g(x)) = 0$, where $f(x) = x^2 - 4$ and $g(x) = x + 2$
(A) -2
(B) 0
(C) 2
(D) -4
(E) 4
- Q7.** Given $f(x) = \frac{1}{x}$ and $g(x) = x^2 - 9$, evaluate $f(g(3))$
(A) $\frac{1}{0}$
(B) $\frac{1}{9}$
(C) $\frac{1}{-8}$
(D) $\frac{1}{-4}$
(E) 0
- Q8.** Determine the domain of $f(g(x))$ where $f(x) = \sqrt{x}$ and $g(x) = x^2 - 16$
(A) $[-4, 4]$
(B) $[-4, \infty)$
(C) $[0, \infty)$
(D) $[-\infty, -4] \cup [4, \infty)$
(E) $[4, \infty)$

Open Ended Questions (FRQ)

- Q9.** Evaluate $f(g(x))$ and determine its domain, given $f(x) = \frac{x}{x+1}$ and $g(x) = x^2 - 2x$. Show all work and explain your reasoning.
- Q10.** Interpret $f(g(x))$ in the context of an electrical circuit, where $f(x) = 2x$ and $g(x) = x + 5$. Provide a sketch of the circuit and explain how $f(g(x))$ relates to the circuit's behavior. Show all work and explain your reasoning.

Chef's Tips

- Tip 1: Encourage students to use graphs and tables to visualize composite functions
- Tip 2: Remind students to check for domain restrictions when evaluating composite functions
- Tip 3: Suggest using real-world examples to illustrate the importance of composite functions